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Genome Note

Hybrid genome assembly and annotation of multidrug-resistant *Staphylococcus aureus* genotype ST672-SCCmec type IVd (2B)Jiffy John^{a,b}, Lekshmi Narendrakumar^c, Sabu Thomas^c, Shijulal Nelson-Sathi^{a,*}^a Bioinformatics Laboratory, Transdisciplinary Biology, Rajiv Gandhi Centre for Biotechnology (RGCB), Thiruvananthapuram, India^b Manipal Academy of Higher Education (MAHE), Manipal, India^c Department of Pathogen Biology, Rajiv Gandhi Centre for Biotechnology, Thiruvananthapuram, India

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ABSTRACT

Objectives: The emergence of multidrug-resistant *Staphylococcus aureus* strains is mainly mediated by mobile genetic elements, such as Staphylococcal Cassette Chromosome *mec* (SCCmec). Currently, SCCmec elements in *S. aureus* are classified into 15 types, with type IV being the most common in hospital and community-associated methicillin-resistant *S. aureus*. Among different subtypes of SCCmec type IV strains (IVa–IVn), the complete genome sequence of the SCCmec IVd (2B) subtype is still lacking. Here, we report the complete genome sequence of multidrug-resistant *S. aureus* SCCmec typeIVd (2B) isolate, *S. aureus* S145.

Methods: *Staphylococcus aureus* S145 was subjected to phenotypic and genotypic characterization. The whole-genome sequencing of S145 was performed using a hybrid-genome approach. The antibiotic-resistance genes were detected and compared with 112 publicly available *S. aureus* genomes.

Results: We obtained a complete genome of S145 with 2.7 Mbp length, Guanine-Cytosine (GC) content of 32.8%, and 2,548 protein-coding regions with 79 virulence factors and 90 antibiotic resistance genes. The S145 has ~17-kb SCCmec, which encodes genes such as *mecA*, *mecR1*, *ccrA2B2*, and SCCmec IVd (2B) subtype gene *CG002*. We detected a ~30-kb multidrug-resistant plasmid with eight antibiotic-resistant genes forming three clusters. Cluster1 encoded for penicillin (*bla1-blaZ-blaR1*), Cluster2 for aminoglycoside-streptothricin (*aph(3')-IIIa-sat4-ΔANT(6)-Ia*), and Cluster3 for macrolides (*msr(A)-mph(C)*) resistance genes. Comparative analysis of Cluster1–Cluster3 revealed that the genetic organization of these clusters resembles resistance genes present in plasmids of USA300 *S. aureus* SCCmec type IVa strains.

Conclusion: Here, we report the complete genome sequence of *S. aureus* SCCmec IVd (2B) that can be used as a reference genome for further comparative genomic analysis.

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1. Introduction

Staphylococcus aureus is one of the most concerning human pathogens because of its ability to cause various skin and soft tissue infections and its potential to develop antibiotic resistance against multiple antibiotics [1]. The accumulation and dissemination of antibiotic resistance genes in the pathogen are mediated by mobile genetic elements such as plasmids, prophages, transposons, cassettes, etc. One significant example is the Staphylococcal Cas-

sette Chromosome *mec* (SCCmec), which carries a methicillin-resistant gene (*mecA*), and a cassette chromosome recombinase gene complex (*ccrA/ccrB/ccrC*) [2]. SCCmec cassettes have been characterized into types and subtypes based on their highly diverse structural organization and genetic content. To date, fifteen SCCmec types (I–XV) have been identified based on the unique combination of *mec* and *ccr* gene complexes, and its variation with the joining (J) region determines the subtypes of each SCCmec type [3]. *Staphylococcus aureus* strains with SCCmec type IV is one of the most significant and challenging types to characterize, as it is the shortest and most mobile structural type of SCCmec. SCCmec type IV is also more variable than other SCCmec types and has different subtypes (subtypes: IVa–IVn) [4]. The complete genome sequences, antibiotic susceptibility, virulence profile, and genomic diversity of

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S. aureus with SCCmec type IVd (2B) subtype is lacking. This study aims to report a complete genome of *S. aureus* with SCCmec type IVd (2B) subtype (S145 hereafter) by hybrid-genome approach using the sequencing reads from Nanopore and Illumina platforms to characterize the genome through annotation and comparative genomics.

2. Materials and methods

Staphylococcus aureus S145 was isolated in the year 2016 from the pus sample of a male patient from a tertiary health care centre in Kerala. Species identification was confirmed by 16S rRNA gene sequencing (Applied BioSystems, Foster City, California) and detecting the *nuc* (nuclease) and *mecA* (methicillin resistance gene) genes. The antibiotic resistance profile was determined by the disc diffusion assay using different antimicrobial agents. *Staphylococcus aureus* ATCC 25923 was used for internal quality control. Experiments were done in duplicates and the inhibition zone diameters were measured and the mean of inhibition zone diameters were compared with NCCLS Disc Diffusion breakpoints [5]. Total genomic DNA of selected multidrug-resistant *S. aureus* was isolated using the Wizard genomic DNA purification kit (Promega), as per the manufacturer's protocol and quantified with a Qubit® 2.0 Fluorometer (Thermo Fisher Scientific, Waltham, MA). To get the complete genome, we pursued a hybrid-genome approach using long-read Nanopore MinION and short-read Illumina MiSeq sequencing. Both the resulting reads were quality checked and quality trimmed using FastQC and Trim Galore (<https://www.bioinformatics.babraham.ac.uk/projects/>). After assembly and extraction of plasmids using PlasmidSPAdes (<https://cab.spbu.ru/software/plasmid-spades/>), finally de novo assembly was conducted using Unicycler v0.8.4.0 (<https://github.com/rrwick/Unicycler>). The complete genome sequences were annotated using the RAST server (<https://rast.nmpdr.org/>). The conventional seven gene multilocus sequence typing (MLST) predictions made use of the *S. aureus* database in PubMLST (<https://pubmlst.org/>). Tools available from the Center for Genomic Epidemiology server (<http://www.genomicepidemiology.org/>) were used to predict the mobile genetic elements and SCCmec typing of the isolate. Resistomes of S145 were predicted using BLAST 2.10.0 (<https://blast.ncbi.nlm.nih.gov/Blast.cgi>) search against an in-house resistance database created by combining resistance genes from Comprehensive Antibiotic Resistance Database (CARD; <https://card.mcmaster.ca/>) and Resfinder (<https://cge.cbs.dtu.dk/services/ResFinder/>). While virulence factors were predicted using BLAST against the curated core dataset (3257 nucleotide sequences) of Virulence Factor Database (VFDB; www.mgc.ac.cn/cgi-bin/VFs/). The resistance and virulence genes selections were set to $\geq 95\%$ of identity over $\geq 90\%$ query coverage and E-values $< 1e-05$ were considered. For comparative genomic analysis of previously reported completely sequenced reference/representative *S. aureus* strains with SCCmec type IV, such as *S. aureus* MW2 (SCCmec type IVa), *S. aureus* HO 5096 0412 (SCCmec type IVh), *S. aureus* VB31683 (SCCmec type IVc: India), and *S. aureus* AMRF4 (SCCmec type IVc: India), we used the progressive mauve genome alignment algorithm (<https://darlinglab.org/mauve/mauve.html>) with default parameters. BLAST Ring Image Generator (<https://brig.sourceforge.net/>) was used to show a genome-wide visualization of coding sequences identity between the newly sequenced S145 genome and those genomes of the *S. aureus* strains mentioned above. Thirty-six plasmid encoded genes annotated using RAST server were compared using blastp program against 112 *S. aureus* SCCmec IV gene sequences, and the hits with $\geq 95\%$ of sequence identity, $\geq 90\%$ query coverage, and an E-value $< 1e-05$ were considered. The heatmap was generated using GraphPad Prism version 9 (www.graphpad.com).

3. Results and discussion

Using a hybrid-genome sequencing approach a total of 2,551,570 Illumina high-quality reads and 17,902 Nanopore long-reads were generated. From the high-quality assembly of S145, two contigs were generated with a total length of 2,747,692 bp and an N50 length of 2,611,360 bp. Also, a plasmid of 29,177 bp was obtained (Supplementary Table S1). Genome annotation resulted in 2,548 protein-coding genes in S145 with a Guanine-Cytosine (GC) content of 32.9%. The S145 consists of 79 RNAs (60 tRNA and 19 rRNA [5S (7), 16S (6), and 23S (6)]) and 4 non-coding RNAs. The plasmid contains eight repeat regions and 36 gene sequences with a GC content of 29.9%. RAST SEED sub-system characterisation classified 2,548 Open Reading Frame (ORFs) in S145 into ORFs that were responsible for amino acid synthesis; carbohydrates synthesis; protein metabolism and virulence; and disease and defence class (Supplementary Fig. S1). The *insilico* multilocus sequence typing analysis using seven housekeeping genes showed that S145 belonged to sequence type (ST) 672. The allelic differences of the housekeeping genes to the reference genes are *arcC* (4), *aroE* (3), *glpF* (1), *gmk* (1), *pta* (11), *tpi* (72), and *yqiL* (11).

Among 2,548 protein-coding genes, we identified 79 virulence genes in S145. S145 was identified to possess staphylokinase (*sak*), staphylococcal complement inhibitor (*scn*), γ -haemolysin (*hlgA/hlgB/hlgC*), leukotoxin ED (*lukE/lukD*), and enterotoxins (*sed*, *sej*, *sek*, *seq*, and *ser*). Biofilm formation-related genes (*icaABCD*) and aureolysin (*aur*), which are important in pathogenesis, were also present (Supplementary Table S2). Although the prevalence of Panton-Valentine leukocidin (*pvl*) gene is low in nosocomial MRSA isolates, the clinical isolate, S145, was identified as a *pvl*-positive MRSA verified by the presence of *pvl* constituting two exoproteins, namely LukS-PV and LukF-PV. A whole-genome comparison of S145 assembly against four representative strains of *S. aureus* with SCCmec type IV subtypes revealed that three staphylococcal pathogenicity islands (SaPIs) and some virulence genes, such as leukotoxin (*lukE* and *lukD*), are uniquely present in S145 (Supplementary Fig. S2).

Phenotypic susceptibility testing revealed that the S145 is resistant to different antibiotics, such as cefoxitin, ciprofloxacin, oxacillin, kanamycin, erythromycin, penicillin G, azithromycin, methicillin, and vancomycin; however, it is susceptible to tobramycin, tetracycline, chloramphenicol, linezolid, and clindamycin (Supplementary Table S1). We found that the S145 chromosome carries a ~17-kb region encoded with methicillin-resistance gene, *mecA*, and its regulatory gene, *mecR1*, and is referred to as SCCmec cassette. Physically linked structure of genes, such as *mecA*, *delta-mecR1*, *IS1272*, *ccrA2B2*, and SCCmec type IVd (2B) subtype gene *CG002*, revealed that the SCCmec cassette in S145 corresponds to SCCmec type IVd (2B) subtype and is very similar to the SCCmec genes structure of reference strain *S. aureus* JCS4469 (AB097677). Further analysis revealed 82 chromosomally encoded antibiotic resistance genes (Supplementary Fig. S2) in addition to the resistance genes linked to the SCCmec cassette in S145. These gene products are involved in different resistance mechanisms, such as antibiotic efflux, antibiotic inactivation, antibiotic target alteration, antibiotic target protection, antibiotic target replacement, etc. (Supplementary Table S3). Detailed analysis of S145 assembly revealed an extra-chromosomal contig with ~30-kb long. Gene annotation revealed that eight antibiotic resistance genes are present in this S145 plasmid and they were found to be grouped in three distinct clusters, namely: penicillin resistant gene cluster 1 (*blaI-blaR1-blaZ*), aminoglycoside-streptothricin resistant gene cluster 2 (*aph(3')-IIIa-sat4-ΔANT(6)-Ia*), and macrolides resistant gene cluster 3 (*msr(A)-mph(C)*). The resistance gene clusters were identified to be flanked with IS257 and Tn552 transposons (Fig. 1). Along with antibiotic resistance genes, a cadmium resistance gene, *cadD*, and its regula-

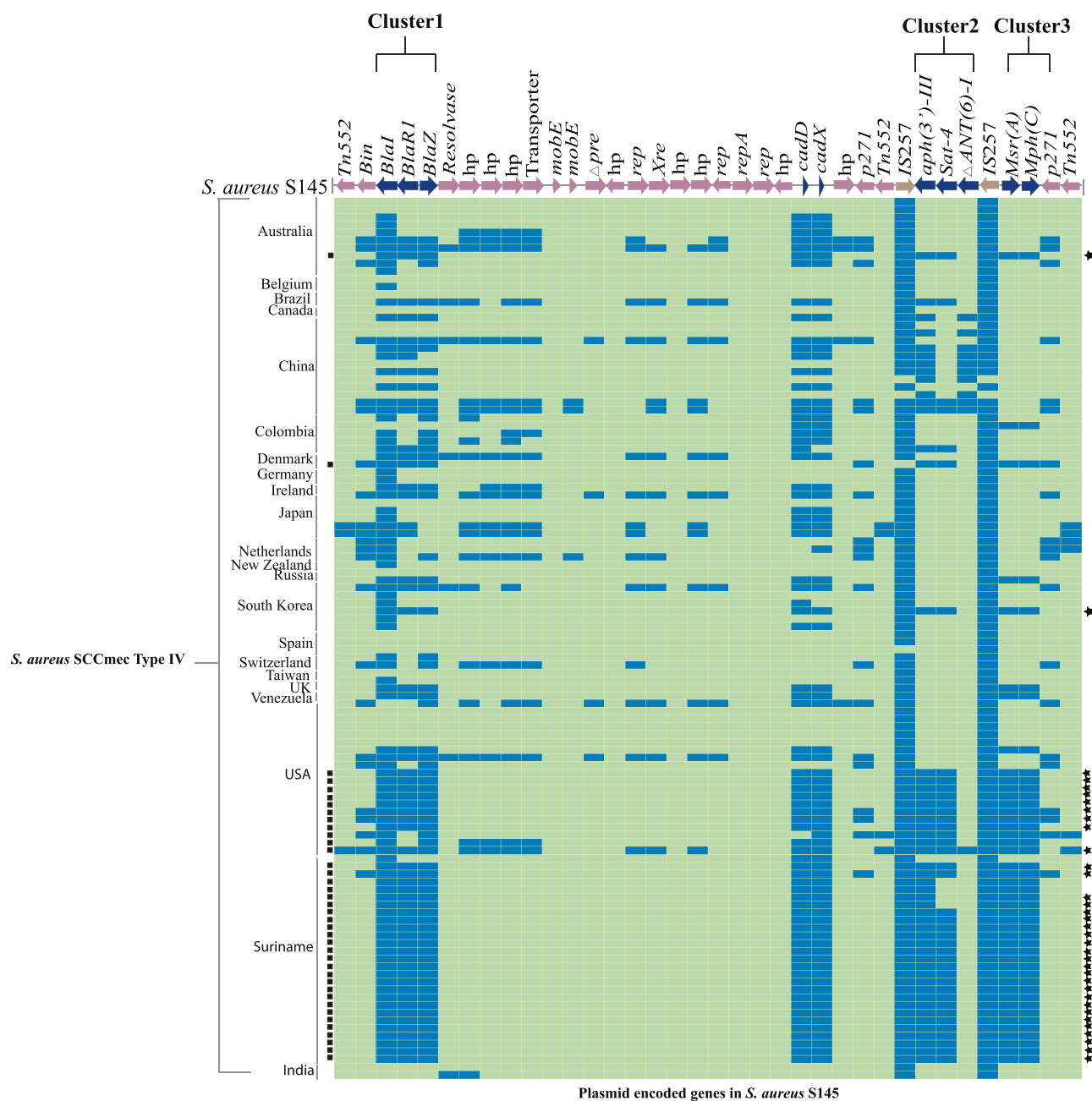


Fig. 1. Distribution of 36 plasmid-encoded genes of *Staphylococcus aureus* S145 in 112 *S. aureus* SCCmec IV strains reported worldwide. X-axis represents the plasmid genes and Y-axis represents the 112 *S. aureus* strains sorted based on the respective countries. The first row represents the 36 annotated genes of S145 plasmid which are shown as arrows with direction representing the + and – strands. Insertion sequence elements are indicated in yellow; plasmid replication and its regulatory gene, transposons, and hypothetical genes are depicted as purple; and the resistance genes (antibiotic and metal) are represented in blue. Inside the matrix, the presence of genes is represented by blue and their absence by green. The genes encoded within the plasmids of reported genomes are highlighted with an asterisk (*). Multidrug-resistance plasmid with eight antibiotic-resistance genes forming three distinct gene clusters, designated as Cluster1 (penicillin), Cluster2 (aminoglycoside-streptothricin), and Cluster3 (macrolides). The *S. aureus* genomes have the same genetic organization as Cluster1 to Cluster3 gene clusters and are represented here as black rectangular boxes.

tor, *cadX*, were present downstream from penicillin resistant gene cluster 1. In S145 plasmid, the aminoglycoside-streptothricin resistant gene cluster 2 (*aph(3')-IIIa-sat4-ΔANT(6)-Ia*) coded for aminoglycoside modifying enzymes and exhibited resistance to gentamycin, streptothricin, and kanamycin; the macrolides resistant gene cluster 3 (*msr(A)-mph(C)*) coded for a ribosomal protection protein and macrolide phosphotransferases, which code resistance to macrolide antibiotics. Gene distribution pattern and linkage analysis of resistance gene clusters in S145 plasmid exhibited a structural resemblance with resistance genes encoded in pUSA300HOUR-like plasmids. Comparative analysis of Cluster1 to Cluster3 of S145 with 112 publicly available *S. aureus* strains re-

vealed that the genetic organization of these three resistant clusters resembles resistance genes present in plasmids of USA300 *S. aureus* SCCmec typeIVa strains (Fig. 1). A recent study from Steinig et al. [6] identified the presence of the same three distinct resistance gene clusters in *S. aureus* ST772 SCCmec V strains; it was reported from India as a chromosomally integrated resistance plasmid. Our study suggests that mobile genetic elements associated with resistance genes might promote the spreading of multidrug resistance in *S. aureus* strains with different SCCmec subtypes. The findings show that continuous complete genomic characterization will aid in understanding the evolution of antibiotic resistance and the surveillance of *S. aureus* strains.

Nucleotide sequence accession numbers

The genomic data of this study were deposited publicly in DDBJ/ENA/GenBank® under Accession: JAJTJT000000000, BioProject: PRJNA791468 and Biosample: SAMN24290962.

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Competing interests

None declared.

Ethical approval

Institutional Biosafety Committee (IBSC) approval was obtained from Rajiv Gandhi Centre for Biotechnology, Thiruvananthapuram, India.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.jgar.2022.12.011](https://doi.org/10.1016/j.jgar.2022.12.011).

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